

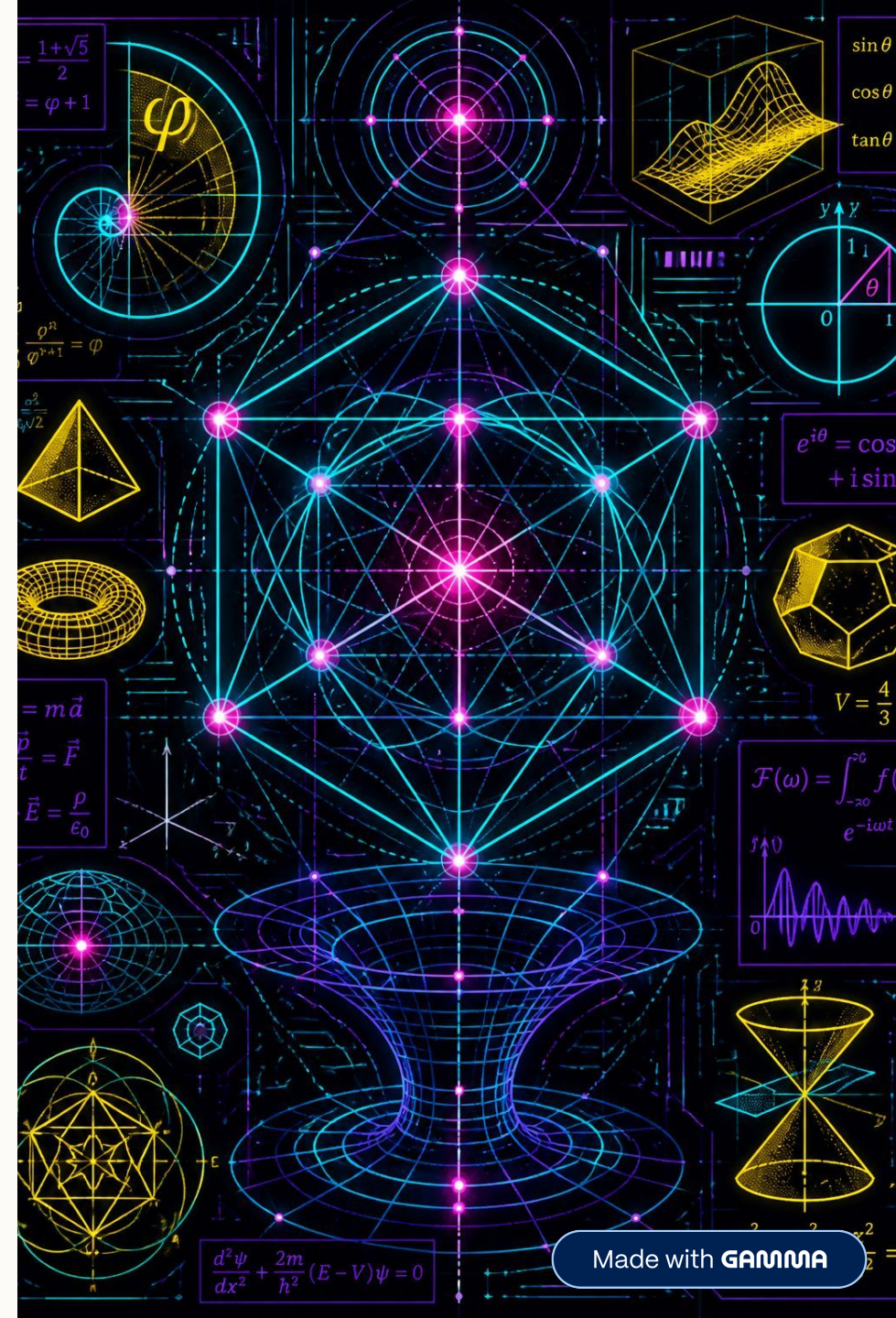
Review of Linear Algebra

A concise journey through vectors, matrices, and the mathematical structures that power modern computing, AI, and engineering.

MATHEMATICS

COMPUTER SCIENCE

ENGINEERING



What is Linear Algebra?

Linear Algebra is the branch of mathematics that studies **vectors**, **matrices**, and **linear equations**. It is the language of structure — providing the tools to describe and manipulate data in high-dimensional spaces.

Computer Science

Algorithms, graphics, and data structures

Artificial Intelligence

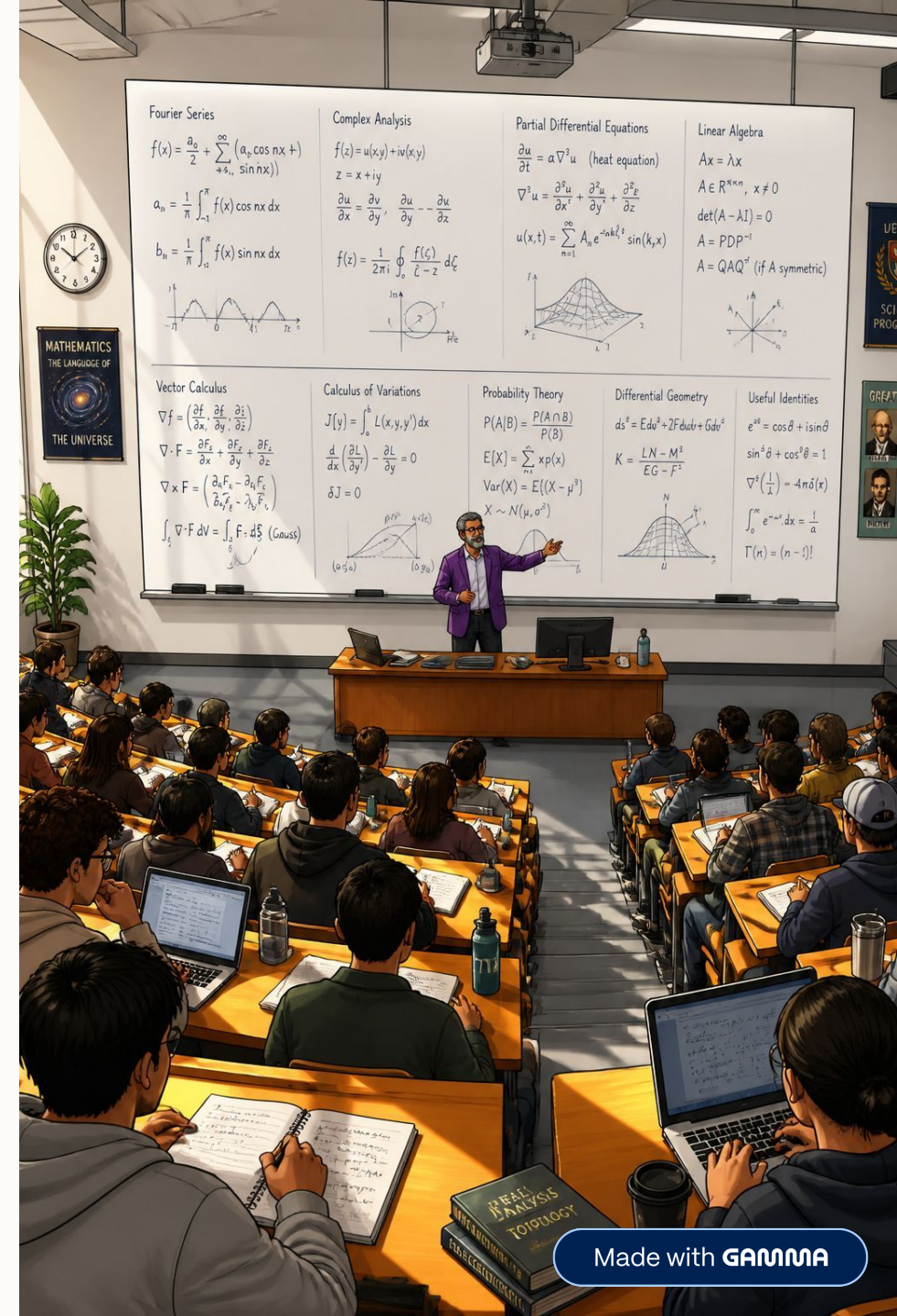
Neural networks and model training

Data Science

Dimensionality reduction and statistics

Quantum Computing

State vectors and unitary operators



Scalars & Vectors

Scalar

A quantity with **magnitude only** — a single real number.

- Temperature: 98.6°F
- Mass: 5 kg
- Time: 12 seconds

Vector

A quantity with **magnitude and direction** — an ordered list of numbers.

- Velocity: speed + heading
- Force: strength + direction
- Displacement: distance + orientation

Example: $\mathbf{v} = (3, 4)$

Magnitude: $|\mathbf{v}| = \sqrt{3^2 + 4^2} = 5$

Vector Operations

Three fundamental operations form the backbone of vector algebra. Each has a clear geometric interpretation and wide practical use.

Vector Addition

Add corresponding components.
Geometrically, this places vectors tip-to-tail.

$$(2, 3) + (4, 1) = (6, 4)$$

Scalar Multiplication

Multiply every component by a scalar, scaling the vector's length.

$$3 \cdot (2, 1) = (6, 3)$$

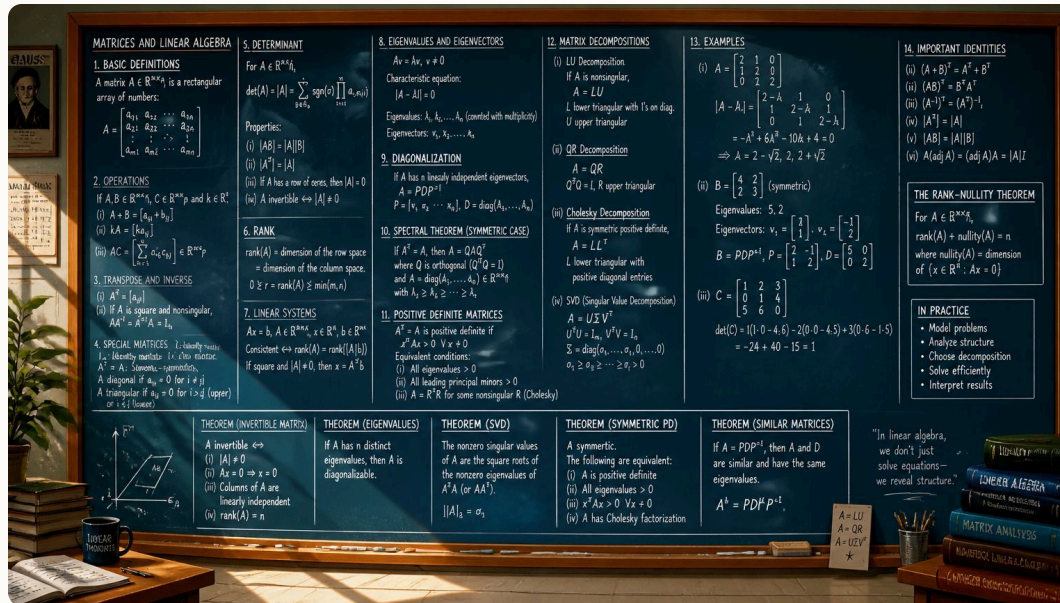
Dot Product

Multiply corresponding components and sum. Measures alignment between vectors.

$$(2, 3) \cdot (4, 5) = 8 + 15 = 23$$

❏ Key properties: dot product is **commutative** and **distributive**; vector addition is **associative**.

Matrices



A **matrix** is a rectangular arrangement of numbers organized into rows and columns. Matrices encode linear transformations and systems of equations compactly.

Example 2×2 matrix:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Row
Single row

Column
Single column

Square
 $n \times n$

Identity
Diagonal 1s

Zero
All zeros

Diagonal
Off-diag zeros

Matrix Operations



Addition & Subtraction

Add or subtract corresponding entries. Matrices must share the same dimensions.



Matrix Multiplication

Dot product of rows with columns. Requires compatible inner dimensions.



Scalar Multiplication

Multiply every entry by a scalar value, uniformly scaling the matrix.



Transpose

Flip rows into columns. Example: $A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$

Determinant & Inverse

Determinant

For a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$:

$$\det(A) = ad - bc$$

The determinant is a single scalar that reveals whether a matrix is invertible and how it scales area or volume.

Matrix Inverse

The inverse A^{-1} exists **only if** $\det(A) \neq 0$.

It satisfies:

$$A \cdot A^{-1} = I$$

Inverses are used to solve linear systems of the form $A\mathbf{x} = \mathbf{b}$ by computing $\mathbf{x} = A^{-1}\mathbf{b}$.

Systems of Linear Equations

A system of linear equations can be written compactly as $A\mathbf{x} = \mathbf{b}$, where A is the coefficient matrix, $\mathbf{x}$ is the unknown vector, and $\mathbf{b}$ is the result vector.



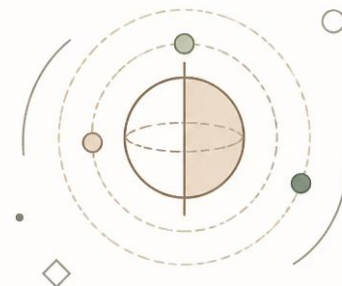
Each method has trade-offs in efficiency and numerical stability. **Gaussian Elimination** is the workhorse of computational linear algebra.

Eigenvalues & Eigenvectors

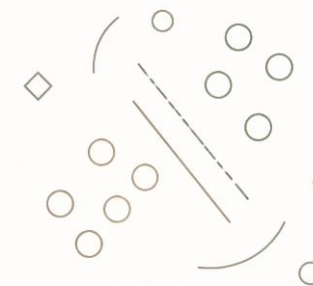
An **eigenvector** $\mathbf{x}$ of a matrix A is a special vector that does not change direction when transformed — it is only scaled by its **eigenvalue** λ :

$$A\mathbf{x} = \lambda\mathbf{x}$$

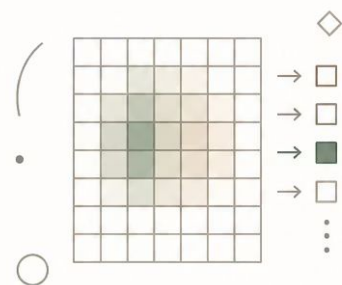
Eigenvectors reveal the intrinsic "axes" of a transformation. Eigenvalues tell us how much stretching or shrinking occurs along each axis.



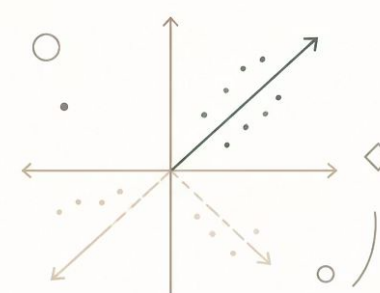
Quantum Computing
(energy states and observables)



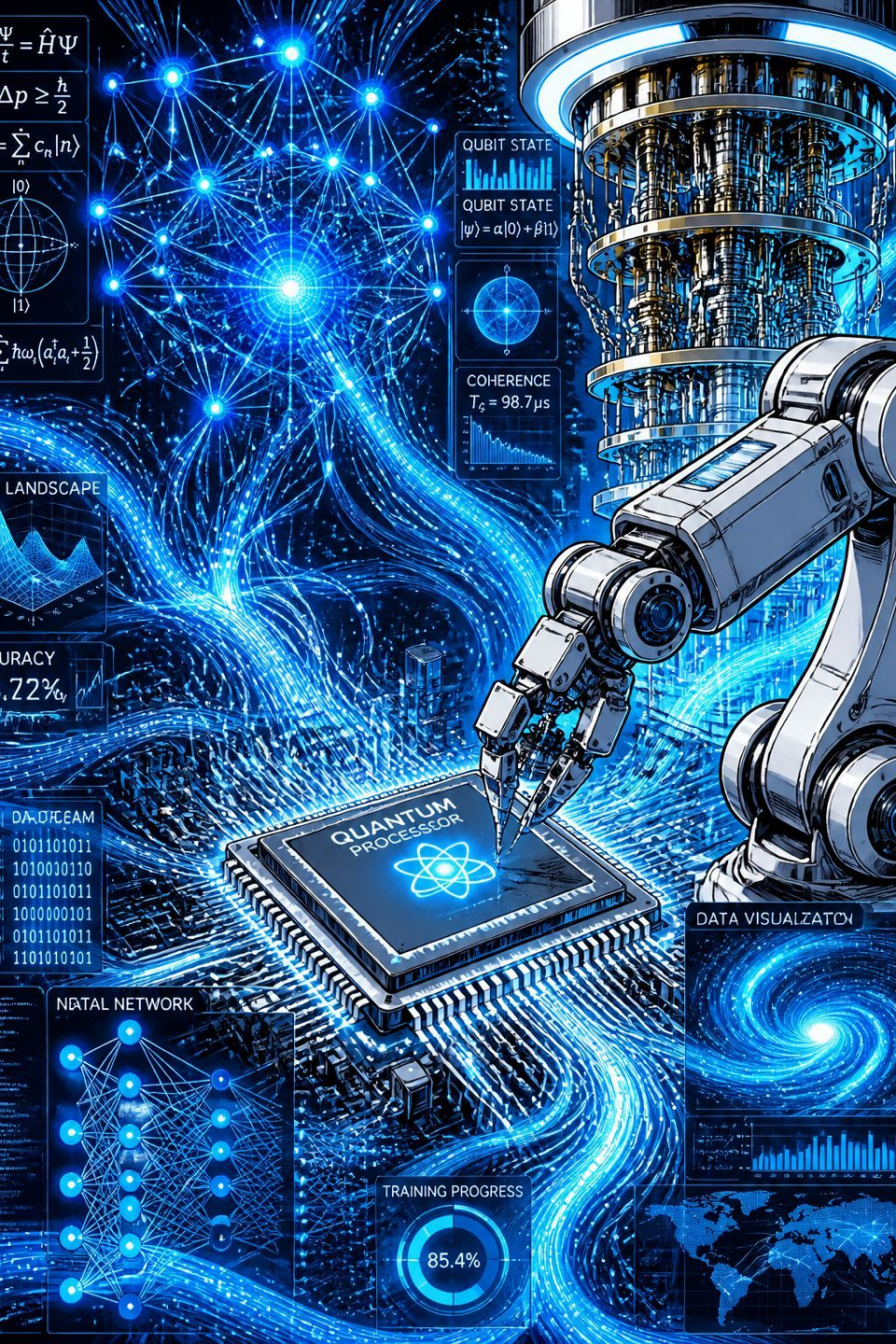
Machine Learning
(feature extraction and optimization)



Face Recognition
(eigenfaces for dimensionality reduction)



PCA - Principal Component Analysis
(data compression and visualization)



Why Linear Algebra Matters

Vectors & Matrices

The fundamental data structures of computation

Operations & Inverses

Tools for transforming and solving systems

Eigenanalysis

Unlocking patterns in data and physics

Modern Applications

AI, quantum computing, graphics, and beyond

- 📌 **Key Takeaway:** Linear Algebra is not just abstract math — it is the operating system of modern technology. Every neural network, physics simulation, and recommendation engine runs on these foundations.